

The critical local free log-Sobolev inequality

A candidate full solution to Open Problem 1 of arXiv:1209.2970

Automated mathematical discovery run `fa_banach_001`
Lane 16, model GPT5.6

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Status. Candidate full solution, likely valid, subject to expert review. The proof is analytic and has no computational dependency.

1 Source problem and answer

Popescu’s *Local Functional Inequalities in One Dimensional Free Probability* [1] defines, for probability measures on $[-2, 2]$,

$$\mathcal{H}(\mu, \nu) = - \iint \log |x - y| d(\mu - \nu)(x) d(\mu - \nu)(y),$$

the semicircle law

$$d\alpha(x) = \frac{\sqrt{4 - x^2}}{2\pi} dx,$$

and the Hilbert transform

$$H\sigma(x) = \text{p. v.} \int \frac{2}{x - y} d\sigma(y).$$

Open Problem 1 in Section 6 asks whether there is a universal $C > 0$ such that

$$C\mathcal{H}(\mu, \nu) \leq \left(\int_{-2}^2 |H\mu - H\nu|^{3/2} d\alpha \right)^{4/3}. \quad (1)$$

The paper proves that no analogous inequality holds when $p < 3/2$ and asks, in Open Problem 2, for the smallest valid $p \in (3/2, 2)$ if the endpoint fails. Figure 1 reproduces the source statement and its equivalent sine-series formulation.

Theorem 1 (Endpoint and optimal exponent). *There is a universal constant $C_* > 0$ such that (1) holds for all probability measures μ, ν on $[-2, 2]$ (with the usual extended-value interpretation). Consequently the smallest exponent in the source’s equation (6.1) is exactly $p = 3/2$.*

2 Proof intuition

The singular weight in the source’s trigonometric reformulation is not an obstruction; it is a conformal Jacobian. Weighted duality changes the $p = 3/2$ estimate into

$$\|G\|_{L^3((0,\pi), dt/\sin t)} \lesssim \left(\sum_{k \geq 1} k |b_k|^2 \right)^{1/2}, \quad G(t) = \sum b_k \sin kt.$$

$$= \frac{\eta^{3/2}}{8}(-4 + 6 \log(2) - 2 \log(1 - r)) + O((\log(1 - r))^2)$$

which does not rule out (6.1).

For $p = 3/2$ we do not have a counterexample to (6.1) nor a proof of validity of it. We post this as an open problem here.

Open Problem 1. *There is a constant $C > 0$ such that for any probability measures μ, ν on $[-2, 2]$,*

$$C\mathcal{H}(\mu, \nu) \leq \left(\int |H\mu - H\nu|^{3/2} d\alpha \right)^{4/3}.$$

A positive answer to this question would give the optimal p for which (6.1) is true. A negative answer would continue with the following.

Open Problem 2. *Is there a $3/2 < p < 2$ such that for some constant $C_p > 0$, (6.1) holds true for any probability measures μ, ν ? And if so, what is the smallest such p ?*

A reformulation of these open problems in terms of trigonometric series can be done based on (5.6), (5.8) and the definition of the Chebyshev polynomials of second kind. Using an approximation of the measures μ, ν with measures of the form $\phi d\beta$ and $\psi d\beta$, an equivalent form of (6.1) is the following. What is the smallest $1 < p$ such that for any $n \geq 1$ and $a_1, a_2, \dots, a_n \in \mathbb{R}$,

$$(6.2) \quad C_p \sum_{k=1}^n \frac{a_k^2}{k} \leq \left(\int_0^\pi \left| \sum_{k=1}^n a_k \sin(kt) \right|^p \sin^{2-p}(t) dt \right)^{2/p} ?$$

The conclusion of this section is that definitely $3/2 \leq p$, but it is not clear that the smallest p is exactly $3/2$.

Figure 1: Open Problems 1–2 and equation (6.2), source paper p. 16.

Extend G harmonically to the upper half-disk. Its Dirichlet energy is exactly $(\pi/2) \sum k|b_k|^2$. The conformal map

$$z \mapsto \log \frac{1+z}{1-z}$$

sends the half-disk to a strip, the diameter (where the sine extension vanishes) to the strip's bottom, and the semicircle to its top with $|dx| = dt/\sin t$. On a fixed strip, the bottom Dirichlet condition makes the top trace an inhomogeneous $H^{1/2}(\mathbb{R})$ function, hence an $L^3(\mathbb{R})$ function. This proves the dual estimate. The Chebyshev/logarithmic-potential identities in the source then carry it back to arbitrary measures.

3 The critical weighted trace estimate

Lemma 1 (Half-disk/strip trace). *For every finite real sine polynomial $G(t) = \sum_{k=1}^n b_k \sin(kt)$,*

$$\left(\int_0^\pi |G(t)|^3 \frac{dt}{\sin t} \right)^{1/3} \leq C \left(\sum_{k=1}^n k b_k^2 \right)^{1/2}. \quad (2)$$

Proof. On the upper half-disk $D_+ = \{re^{it} : 0 < r < 1, 0 < t < \pi\}$, set

$$u(r, t) = \sum_{k=1}^n b_k r^k \sin(kt).$$

Orthogonality and a direct polar-coordinate calculation give

$$\int_{D_+} |\nabla u|^2, dA = \frac{\pi}{2} \sum_{k=1}^n k b_k^2. \quad (3)$$

Indeed, the k th radial and angular terms together contribute $\pi k^2 \int_0^1 r^{2k-1} dr = \pi k/2$.

The map

$$\Phi(z) = \log \frac{1+z}{1-z}$$

maps D_+ conformally onto $S = \mathbb{R} \times (0, a)$, $a = \pi/2$. It sends the diameter to $y = 0$. On the semicircle,

$$\Phi(e^{it}) = \log \cot(t/2) + i\pi/2, \quad \left| \frac{d}{dt} \log \cot(t/2) \right| = \frac{1}{\sin t}.$$

Let $v = u \circ \Phi^{-1}$. Conformal invariance of planar Dirichlet energy and (3) show that $v = 0$ on $y = 0$ and

$$\int_S |\nabla v|^2 = \frac{\pi}{2} \sum_{k=1}^n k b_k^2.$$

Its top trace $h(x) = v(x, a)$ satisfies $h(\log \cot(t/2)) = G(t)$.

For completeness, we prove the needed strip trace bound. Fourier transform in x . Among functions V with $V(0) = 0$ and $V(a) = \widehat{h}(\xi)$, the minimum of

$$\int_0^a (|V'(y)|^2 + \xi^2 |V(y)|^2) dy$$

is $|\xi| \coth(a|\xi|) |\widehat{h}(\xi)|^2$, with limiting value $a^{-1} |\widehat{h}(0)|^2$ at zero. Hence, up to Fourier-normalization constants,

$$\int_{\mathbb{R}} (1 + \xi^2)^{1/2} |\widehat{h}(\xi)|^2 d\xi \leq C_a \int_S |\nabla v|^2. \quad (4)$$

Hausdorff–Young followed by Holder gives

$$\|h\|_3 \leq C \|\widehat{h}\|_{3/2} \leq C \left(\int_{\mathbb{R}} (1 + \xi^2)^{1/2} |\widehat{h}(\xi)|^2 d\xi \right)^{1/2};$$

the last step uses $\int_{\mathbb{R}} (1 + \xi^2)^{-3/2} d\xi < \infty$. Combining this with (4), conformal invariance, and $dx = dt/\sin t$ proves (2). $\square$

Lemma 2 (Endpoint sine inequality). *For every finite real sequence $(a_k)_{k=1}^n$,*

$$\sum_{k=1}^n \frac{a_k^2}{k} \leq C \left(\int_0^\pi \left| \sum_{k=1}^n a_k \sin(kt) \right|^{3/2} \sin^{1/2} t dt \right)^{4/3}. \quad (5)$$

The same estimate holds, by duality, for every $F \in L^{3/2}((0, \pi), \sin^{1/2} t, dt)$, with the left side interpreted as the weighted square sum of its sine coefficients.

Proof. Write $F = \sum a_k \sin kt$ and $G = \sum b_k \sin kt$. Orthogonality and weighted Holder yield

$$\begin{aligned} \left| \sum_{k=1}^n a_k b_k \right| &= \frac{2}{\pi} \left| \int_0^\pi F(t) G(t) dt \right| \\ &\leq C \left(\int_0^\pi |F|^{3/2} \sin^{1/2} t dt \right)^{2/3} \left(\int_0^\pi |G|^3 \frac{dt}{\sin t} \right)^{1/3} \\ &\leq C \left(\int_0^\pi |F|^{3/2} \sin^{1/2} t dt \right)^{2/3} \left(\sum_{k=1}^n k b_k^2 \right)^{1/2}, \end{aligned}$$

where Lemma 1 gives the last line. Taking the supremum over $\sum k b_k^2 \leq 1$ and squaring proves (5). The same dual argument defines and bounds all finite coefficient truncations for a general weighted $L^{3/2}$ function; monotone convergence of the square sums gives the last assertion. $\square$

4 Passage to probability measures

Proof of Theorem 1. Let $\sigma = \mu - \nu$ and

$$m_k = \int_{-2}^2 T_k(x/2) d\sigma(x), \quad k \geq 1.$$

We recall the standard Chebyshev expansion of the logarithmic kernel. If $x = 2 \cos t$ and $y = 2 \cos s$, then, in Abel-regularized form,

$$-\log |x - y| = 2 \sum_{k \geq 1} \frac{\cos(kt) \cos(ks)}{k}. \quad (6)$$

Constants disappear because $\sigma([-2, 2]) = 0$. Consequently

$$\mathcal{H}(\mu, \nu) = 2 \sum_{k \geq 1} \frac{m_k^2}{k}, \quad (7)$$

with value $+\infty$ when the series diverges. This is also the Chebyshev-energy identity used in the source's derivation of equation (6.2).

Put

$$U(t) = - \int_{-2}^2 \log |2 \cos t - y| d\sigma(y), \quad F(t) = \sin t H\sigma(2 \cos t).$$

Differentiating the Abel-regularized expansion distributionally gives

$$F = U' = -2 \sum_{k \geq 1} m_k \sin(kt). \quad (8)$$

Thus the sine coefficients of F are $a_k = -2m_k$. Also, under $x = 2 \cos t$,

$$\int_{-2}^2 |H\sigma|^{3/2} d\alpha = \frac{2}{\pi} \int_0^\pi |F(t)|^{3/2} \sin^{1/2} t dt. \quad (9)$$

If the right side of (1) is infinite, there is nothing to prove. Otherwise Lemma 2, in its dual form, applies to F . Equations (7)–(9) yield

$$\mathcal{H}(\mu, \nu) = \frac{1}{2} \sum_{k \geq 1} \frac{a_k^2}{k} \leq C \left(\int_{-2}^2 |H\mu - H\nu|^{3/2} d\alpha \right)^{4/3}.$$

This proves Open Problem 1. Popescu proves failure for every $p < 3/2$. Because α is a probability measure, $\|f\|_{L^p(\alpha)}$ is nondecreasing in p ; hence the endpoint also implies all $p \geq 3/2$. The optimal exponent is therefore exactly $3/2$. $\square$

5 Verification, scope, and novelty

The proof is dimension-free and uses no positivity or smooth-density hypothesis beyond what is encoded in the extended-value formulation. Abel regularization justifies (6)–(8) for arbitrary finite signed measure differences; alternatively, one may first prove the claim for smooth arcsine densities exactly as in the source’s equivalence argument and pass to the limit.

A bounded novelty search checked the exact title and arXiv id, quoted text of Open Problem 1, the precise weights in equation (6.2), searches combining “free Log-Sobolev” with “ $3/2$ ”, and the author’s later arXiv:1511.05274. No later paper claiming the endpoint or the conformal-strip argument was found. Novelty confidence is moderate rather than definitive.

The principal review targets are: (i) the zero-frequency limit in the strip multiplier, which is what makes the trace norm inhomogeneous; and (ii) the standard Abel identity (7) under the source’s convention for logarithmic energy. Neither is an unproved dependency.

References

- [1] I. Popescu, *Local Functional Inequalities in One Dimensional Free Probability*, J. Funct. Anal. **264** (2013), no. 6, 1456–1479; [arXiv:1209.2970](#).